

Dossier d'analyse critique

Green LLM Routing



REY LLOPIS Jaime - VASQUEZ Juan Martin - Lou REINA-KUNTZIGER
- DIAZ Juan Jose - SHILI Ibrahim - GAROUX Ewan

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Green LLM

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1. Abstract:

This project proposes *Green LLM*, an intelligent routing system designed to reduce the environmental impact of Large Language Model (LLM) inference. Instead of systematically relying on a single, large, energy-intensive model, the system dynamically selects the most appropriate LLM for each query based on multiple factors, including query context, user profile, organizational structure, expected quality, and estimated carbon footprint.

The approach combines four key components: (1) a carbon footprint estimation model derived from prior AGIR work, (2) a machine learning-based routing system inspired by LLMRouter, (3) a structured multi-level usage model (user, group, enterprise), and (4) a repository of past queries leveraged through similarity-based retrieval to reuse context and reduce redundant computations.

By integrating environmental metrics directly into the decision-making process, the project aims to explore and optimize the trade-off between response quality and carbon emissions. In addition, it introduces explainability and visualization mechanisms to make the environmental impact of AI usage transparent to end users and organizations.

2. Description:

Green LLM is designed as a carbon-aware LLM routing application that optimizes inference efficiency while maintaining acceptable performance levels. The system architecture combines intelligent routing, contextual memory, and carbon impact estimation within a unified framework.

2.1 System Architecture:

The system is structured around several core components:

- **Query Repository & Similarity Module**

A database stores previously processed queries along with their embeddings, selected models, responses, and associated carbon costs. When a new query is received, the system performs similarity-based retrieval to identify related past queries. If relevant matches are found, their context can be reused or injected into the prompt, reducing computational needs and enabling the use of smaller models.

- **LLM Routing Engine**

Inspired by LLMRouter, this module selects the most suitable model for each query. The decision is based on:

- query complexity,

- historical usage patterns,
 - expected quality,
 - estimated carbon footprint.
- **Carbon Estimation Module**
The system integrates a carbon estimation model based on usage metrics such as:
 - number of users,
 - frequency of requests,
 - input/output tokens,
 - model type,
 - emission factors.

A conceptual formulation is:

- total consumption = usage volume × tokens per request
- CO₂ emissions = total consumption × emission factor

This enables real-time estimation of environmental impact at multiple levels (query, user, group, enterprise).

- **Hierarchical Data Model**
The application follows a structured organization:
 - Application → Enterprise → Group → User
This supports both:
 - **B2B usage**, centered on organizations and teams

2.2 Functionalities:

The system provides features at different levels:

- **User level**
 - Display individual CO₂ impact
 - Recommend the most suitable LLM
 - Suggest lower-carbon alternatives for the queries
 - Provide a simple view of personal footprint
- **Group level**
 - Measure CO₂ emissions per group
 - Track request volume and token usage
 - Compare usage patterns across groups
- **Enterprise level**
 - Visualize global CO₂ impact
 - Compare groups
 - Monitor usage and costs
 - Access a consolidated dashboard

2.3 Explainability:

A key aspect of the system is explainability. The router must justify its decisions by providing interpretable outputs such as:

- why a specific model was selected,
- what trade-offs were considered (quality vs carbon),
- what alternative models could have been used.

Example explanations include:

- “Model A was selected as it offers a good balance between quality and low CO₂ emissions.”
- “Model B was not selected due to its high estimated carbon cost.”

2.4 MVP Scope:

The Minimum Viable Product focuses on:

1. the routing engine
2. user management
3. group integration
4. basic enterprise dashboard
5. server-side application and front-end interface

3. Project Objectives:

The main objectives of the project are:

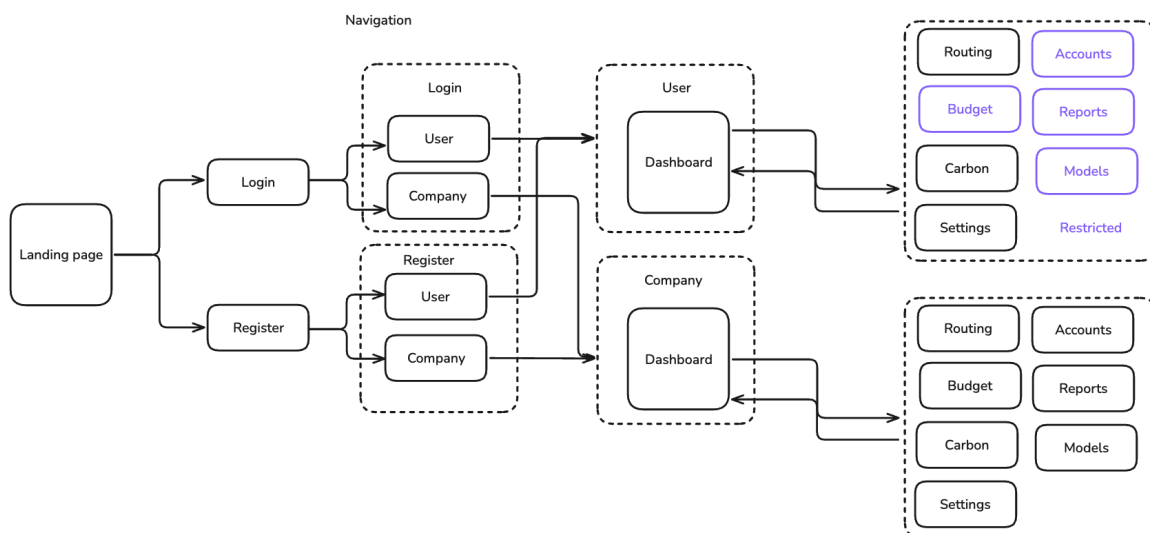
- **Design and implement a carbon-aware LLM routing system**
Capable of dynamically selecting models based on context, performance requirements, and environmental impact
- **Integrate a carbon footprint estimation model**
Based on token usage, model characteristics, and emission factors
- **Build a functional MVP application**
Including routing engine, user/group structure, and basic dashboard
- **Define and structure a hierarchical data model**
Supporting both B2B (enterprise-driven) and B2C (user-centric) use cases
- **Evaluate environmental and performance trade-offs**
By comparing the proposed system to a baseline single-model approach
- **Design relevant monitoring indicators**, including:
 - number of requests
 - token consumption
 - response time
 - estimated CO₂ emissions

- emissions per user/group/enterprise
- selected model and quality metrics
- **Ensure system explainability**
By providing transparent and interpretable routing decisions
- **Provide actionable recommendations**
For building more sustainable and responsible AI systems
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4. Requirements:

4.1 Page navigation:

To effectively address the project requirements, we designed the user interfaces by identifying all the key screens necessary for the proper functioning of the application. We defined interfaces for different types of users, including individual users, groups, and enterprise-level stakeholders.



4.2 UML diagram:

This class diagram represents the core structure of the Green LLM application in a B2B context. It models how the system organizes users, manages interactions with language models, and tracks environmental impact.

The diagram reflects a hierarchical organization where an enterprise is composed of groups, and each group contains users. This structure allows the application to adapt to real-world organizational setups and enables analysis at different levels (individual, group, and company).

At its core, the system revolves around user queries and the corresponding responses generated by language models. These interactions are stored and analyzed to support intelligent routing decisions and to estimate the associated carbon footprint.

Overall, the diagram illustrates how the application connects organizational data, user activity, and LLM usage in order to provide both operational functionality (query handling, model selection) and analytical capabilities (CO₂ impact tracking, usage monitoring, and optimization).

